



The rise and characteristics of China in global environmental governance: Insights from China's global environmental leadership database

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ABSTRACT

With China's progress on green development and recent geopolitical shifts, the international community has high expectations for China to lead global environmental governance. However, China's efforts to lead international environmental cooperation remain understudied. We developed China's Global Environmental Leadership (CGEL) Database to systematically track international environmental governance initiatives led by Chinese actors. Our analysis of 291 initiatives reveals that China has become increasingly active in leading international environmental cooperation, with the geographic scope of these initiatives expanding rapidly. While central government agencies have largely driven this expansion, we observe a trend of diversification with a growing number of Chinese non-state actors developing their own initiatives. On governance functions, China-led initiatives focus primarily on information-sharing, capacity-building, and scientific knowledge production. The CGEL Database offers granular data for further quantitative and qualitative investigations of China's influence on global environmental governance and a transferable methodology for tracking other countries' leadership initiatives.

1. Introduction

Governing planetary challenges such as climate change and biodiversity loss requires collective action across borders and governance levels. With the rise of emerging economies and a polycentric system governing sustainability transitions, new leaders began to emerge in promoting collective action (Chan et al., 2022; Hale, 2020; Mazzega et al., 2025). In this respect, China's ascendancy as a new global power has brought fundamental changes to global sustainability transitions (Finamore, 2018; Watts, 2025). China now holds a pivotal position in global environmental governance as the world's biggest greenhouse gas emitter as well as the most important powerhouse of clean energy technologies (IEA, 2024). Over the last two decades, the country has undergone a significant environmental turn, through strengthened domestic regulations and increasingly proactive action to promote international cooperation (Greenstone et al., 2021; Harlan et al., 2025). Therefore, China's engagement in international environmental cooperation has far-reaching implications for global sustainability, especially in the context of the US' recent retreat from climate action (Sun and Ye, 2025).

To what extent is China becoming a leader in global environmental

governance? And how do different Chinese actors pursue leadership roles in international arenas? These are important questions to understand not only the changing power dynamics in global environmental governance, but also future trajectories of sustainability transitions. However, views on China's shifting role remain debatable among global governance researchers. On one hand, some international relations scholars believe that a rising China challenges the Western-led liberal international order and promotes alternative institutions and norms (Doshi, 2021). On environmental governance, existing research has suggested that through its overseas engagement, China has exported alternative governance approaches such as "coercive environmentalism" (Li and Shapiro, 2020) and "eco-developmentalism" (Zhu et al., 2024) or has projected a "global green vision" combining industrial policy with geopolitical ambitions (Bosi-Moreira, 2025). On the other hand, other researchers underscore that China has benefited from many Western-led global governance systems, and therefore, continuously accepts existing international orders that align with its interests (Johnston, 2019). This argument seems valid in the case of environmental and climate governance where China has a strong domestic interest in supporting existing international institutions and norms (Wallace and Wallace, 2021; Sun and Yu, 2023). Moreover, this debate

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has been largely based on macro-level analysis, instead of granular data on Chinese actors' efforts to lead environmental cooperation so that one cannot fully comprehend China's evolving influence on earth system governance. To address this knowledge gap, we develop an approach to systematically track the multi-faceted efforts of Chinese actors to lead international environmental cooperation.

Our approach employs a broad definition of "global environmental governance" – as "the sum of the formal and informal rule systems and actor networks at all levels of human society that are set up to steer societies towards preventing, mitigating and adapting to environmental change" – to consider initiatives seeking to govern environmental challenges at multilateral, bilateral, and transnational levels (Biermann, 2014, p. 9). We operationalize the concept of "leadership" by focusing on the governance initiatives developed or managed by Chinese actors. This choice reflects the understanding that leadership in global environmental governance is not merely an abstract attribute, but is enacted through concrete actions that mobilize resources, shape norms, or build partnerships (Helms, 2012). Governance initiatives, therefore, serve as an empirical anchor for assessing leadership activities by translating abstract claims into observable practices, such as agenda-setting, resource provision, institution-building, and policy experimentation.

To be qualified as "governance," an initiative must involve an intentional process of steering and must secure recognition of its authority from the individuals and organizations it engages with (Andonova et al., 2009). This definition aligns with a key element in the leadership relationship in terms of being recognized by others (Ahlquist and Levi, 2011). At the same time, the concept of governance broadens the narrow definition of leadership, which implies that those being led must follow the order or solutions suggested by the leader. This is particularly important in a polycentric system governing global environmental change where leadership is mostly exercised through convening, coordinating and pioneering (Tobin et al., 2024). We also unpack "China" to map the diversity of Chinese actors leading international environmental governance initiatives and assess whose leadership is directed towards whom and is exercised with whom. This approach provides a more nuanced understanding of how leadership is practiced.

A key element in our framework concerns the governance functions through which leadership is exercised in practice. While existing literature has identified different forms of leadership in global environmental politics (e.g., structural, entrepreneurial, and intellectual leadership) (Kopra, 2020; Young, 1991), such typologies remain highly abstract and can hardly be operationalized at the activity level. To bridge this gap, we propose six types of governance functions, which directly connect to specific activities aimed at addressing environmental challenges (see Table 1 and the codebook in Supplementary Materials).

Using this approach, we developed China's Global Environmental Leadership (CGEL) Database, which, to our knowledge, is the first and most comprehensive database gathering granular data on China-led cross-border environmental governance initiatives. The database provides an empirical foundation for examining China's environmental

leadership activities in various international arenas. Drawing on descriptive statistics and social network analysis, we assess key trends in the initiatives in the CGEL Database, including the main actors involved and their strategies.

Our findings show that over the past two decades, Chinese actors have become increasingly active in leading cross-border environmental governance, developing 291 initiatives operating across different governance levels and quickly expanding the geographic scope of their leadership activities. Although the Chinese state has been a key driver of many initiatives, we observe a remarkable growth of transnational governance initiatives involving non-state actors. On the focus of China's leadership activities, we find a strong emphasis on capacity-building and scientific knowledge production, which embodies a more practical approach to supporting its international partners. The database thus provides novel insights into the ways in which Chinese actors pursue leadership in global environmental governance and shape sustainability transitions around the world. Beyond these observations, the CGEL Database, by being publicly accessible, will serve as a public good for researchers, policymakers, and other practitioners of environmental governance, which can support a range of further investigations, such as assessing the social and environmental impacts of China-led governance initiatives, identifying cases for more in-depth qualitative analysis of specific initiatives, and also comparing China's environmental leadership with other countries.

2. Methodology

The CGEL Database tracks the international environmental governance initiatives led by Chinese actors in the 21st century. In this section, we explain our methodology to construct this database, including how the data were collected, processed, and analysed. In line with the FAIR (Findable, Accessible, Interoperable, and Reusable) principles, the CGEL Database is openly accessible through the University of Bath Research Data Archive with a registered DOI (<https://doi.org/10.15125/BATH-01664>), as well as in the Supplementary Materials of this article. The data are provided in standard, non-proprietary formats and accompanied by a full codebook documenting all variables and coding methods, enabling other researchers to access, interpret, and reuse the database.

2.1. Scope of the database

The unit of analysis in our database is a governance initiative operating at three governance levels – multilateral, bilateral, and transnational. The initiatives included in our database either primarily address environmental problems or have environmental issues as one of their key concerns. We developed a typology of eleven environmental problems to track the primary and secondary environmental problems addressed by each initiative. For those that address multiple or cross-cutting environmental problems, we coded them as addressing "broad environmental problems". As our interest is in assessing China's leadership activities in global environmental governance, the database only includes the initiatives supporting international cooperation or addressing global problems beyond Chinese borders.

The CGEL Database only includes the initiatives that are either (co-) created by, or their operation is mainly managed by, actors from China to ensure that all cases included are China-led. In our database, we recorded the specific actor(s) (e.g., government agency, industry association, company or civil society organization) who have developed each initiative and their type (e.g., government, business or civil society). For those initiatives that were developed or managed by Chinese actors in collaboration with international actors, we collected data on these international collaborators or co-leads.

Table 1

Governance functions served by China-led international environmental governance initiatives.

Governance function	Explanation
Information sharing and dialogue	Convening stakeholders, facilitating dialogue, building coalitions, fostering cooperation
Capacity-building and training	Enhancing governance capability through training, education, institutional development
Direct action and demonstration	Leading by example through pilots, technology demonstration, and best practices
Provision of funding	Mobilizing financial resources to support governance and incentivize participation
Research and scientific knowledge production	Advancing ideas and solutions via joint research, scientific exchange, and monitoring
Rule-making and standard-setting	Establishing norms, soft rules, technical standards, and best practices

2.2. Data collection

Tracking China-led international environmental governance initiatives is a challenging task due to the lack of official databases of this kind. To overcome this challenge, we developed a robust process to search for prospective cases meeting the inclusion criteria of our database. We followed three steps to identify cases that fit the definition of our database. First, we conducted a series of searches in major existing databases on global environmental governance to identify a list of potential cases that can meet our inclusion criteria (see more details in the search protocol below). Second, to identify cases that are not included in these databases, our team conducted a scoping review of related academic and grey literature – especially studies on China in global environmental governance. Third, we consulted with experts working on related topics and asked them to suggest additional cases. All cases then underwent a screening process by at least two researchers in our team to confirm their eligibility and remove possible duplicates. These efforts allowed us to add additional cases that have not been recorded in any existing databases. In total, 291 cases were included in the first version of the CGEL Database (208 bilateral, 19 multilateral, and 64 transnational initiatives), which covers the initiatives that were active during the period of 2000–2024. Although we cannot claim that these cases are exhaustive, to our knowledge this is the most comprehensive database to track China's leadership activities in global environmental governance.

2.3. Search protocol

We began with our searches in related existing databases. Six credible databases were used as our main sources, which are: PKULaw Database (<https://www.pkulaw.com/>), International Environmental Agreements Database (IEADB) (Mitchell et al., 2020), SDG Actions Platform (<https://sdgs.un.org/partnerships/browse>), Climate Cooperative Initiative Database (C-CID) (Chan et al., 2022), Yearbook of International Organizations (<https://uia.org/ybio/>), and Database of Bilateral Clean Energy and Climate Agreements with China (Lewis, 2023a). These databases were used to search for eligible initiatives operating at multilateral, bilateral, and transnational levels.

Our team conducted a series of searches in these databases to identify a list of potential cases that can meet our inclusion criteria. To efficiently and accurately retrieve potential cases, a senior researcher thoroughly reviewed the methodologies and structures of these databases and developed a set of guidance materials to help research assistants effectively locate relevant entries in each database. All cases then underwent a screening process by at least two research assistants to confirm their eligibility and remove possible duplicates.

To identify cases that are not included in these databases, our team conducted a scoping review of related academic and grey literature – especially studies on China in global environmental governance. We also consulted with experts working on related topics and asked them to suggest additional cases. These efforts allowed us to add additional cases that have not been recorded in any existing databases.

2.4. Coding

To collect data on each initiative, we developed a codebook specifying the variables to be coded or recoded (see the full codebook in the Supplementary Materials) and used it to train a group of coders. After learning the codebook, all coders underwent an exercise to code a small number of the same cases and compare their results to ensure that they fully understood our data collection process and methodology. Upon completion of this training, coders were assigned tasks to collect and code data for the identified cases. Each coder was asked to cross-check relevant information on the initiatives assigned to them from all possible sources available on the Internet, as well as record all sources used.

While some factual information is straightforward, certain variables

in our database require coders' own interpretations based on the codebook. To minimize bias, each case was independently coded by at least two coders. When collecting data, all coders first independently submitted their coding results to an online form. Those having coded the same cases then had meetings with a small review team led by a senior researcher to compare and verify their coding results. When disagreements occurred, coders were asked to check original sources and deliberate on their choices to reach agreements. Any unresolved issues were then further forwarded to the whole team for a more thorough review. All discussions and subsequent decisions made during these meetings were archived for cross-checking. The first version of the CGEL Database includes 29 variables covering main features of each governance initiative meeting our inclusion criteria. The full database is available in the Supplementary Materials.

3. Results

Using the CGEL Database, we use descriptive statistics and social network analysis to show the temporal, geographic, and institutional dynamics of 291 China-led international environmental governance initiatives. The results below provide insights into the evolution and characteristics of China's leadership activities in global environmental governance.

3.1. Expansion over time

Fig. 1 illustrates the cumulative growth of China-led environmental governance initiatives across governance levels and world regions. The overall trend reveals a gradual but steady increase in the efforts of Chinese actors to lead international environmental cooperation from the late 1980s to the early 2000s, followed by a notable surge after 2010. This reflects China's growing engagement in international environmental cooperation across bilateral, multilateral and transnational levels – a trend consistent with observations of China's increasingly active role in global environmental governance over the past three decades (Hilton, 2016; Qi and Dauvergne, 2022). This growth can be partly attributed to China's deepened commitment to advancing environmental governance in response to intensifying ecological crises and shifting national development imperatives (Wang, 2026). Until 2010, China's engagement in global environmental governance was largely characterized by a form of pragmatism – that is, project-based and issue-specific cooperation oriented towards mutual development gains (Feng and McCarthy, 2025; Wang, 2026). Before 2010, bilateral initiatives dominated China's leadership activities (accounting for around 84% of all pre-2010 initiatives), reflecting primarily two complementary aims: addressing transboundary ecological challenges with neighboring countries – most notably Russia (14 initiatives), Japan (9), and Mongolia (3) – and learning technologies and policies from advanced economies, especially Europe and the US (50 initiatives in total).

However, the rapid growth of transnational initiatives involving non-state actors since 2015 marks a new dynamic in China's involvement in global environmental governance. This rise of China-led transnational initiatives corresponds with Beijing's shift toward green development in China's overseas engagement, especially through the Green Belt and Road Initiative (BRI), in the broader context of the country's embracement of "ecological civilization" (Hansen et al., 2018; Rodenbiker, 2023; Sun and Yu, 2023). Accordingly, in the database, almost half of the initiatives developed after 2015 (33 out of 67 initiatives) are transnational initiatives. This suggests a new phase of China's leadership activities in global environmental governance, where Chinese actors – including both state and non-state actors – have become more proactive in establishing cooperation through more flexible, network-based transnational governance incorporating a diverse group of international partners, including intergovernmental organizations, businesses, and civil society organizations (Geng and Lo, 2023; Guttman et al., 2018; Sun, 2022).

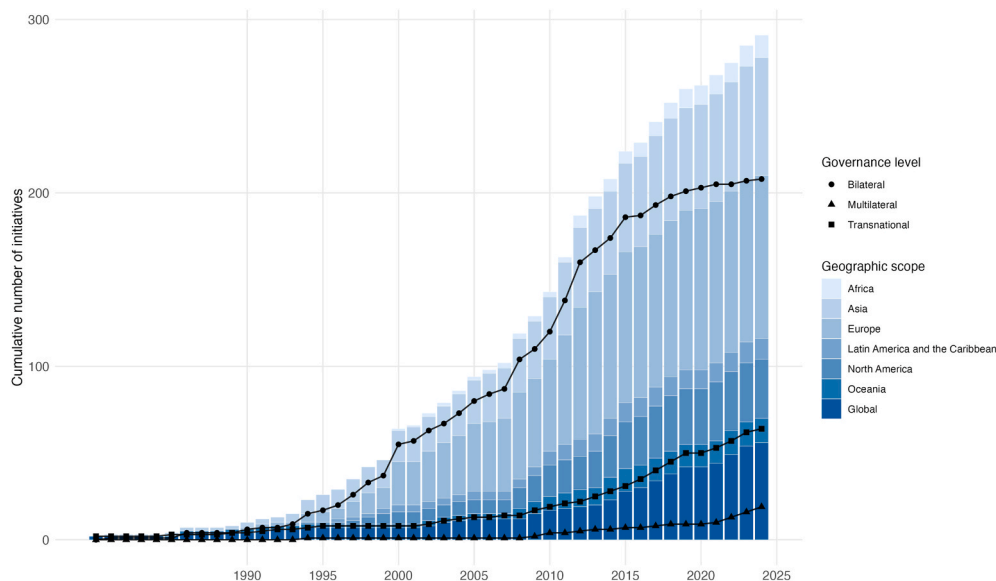


Fig. 1. Temporal evolution of Chinese-led international environmental governance initiatives across governance levels and geographic scope.

Notably, these transnational initiatives increasingly target the environmental footprint of Chinese overseas activities. For example, the Responsible Critical Mineral Initiative, previously Responsible Cobalt Initiative, developed by the China Chamber of Commerce of Metals, Minerals and Chemicals Importers and Exporters, is the first accountability mechanism established by a Chinese industry association to create channels for communities affected by Chinese overseas mineral investments to voice their concerns (Park, 2023). Similarly, the Belt and Road Eco-Industry Cooperation Working Committee, guided by the China Environmental Protection Federation, brings together environmental NGOs and industry actors to address the ecological impacts of Chinese overseas investments along the Belt and Road. Overall, transnational initiatives in our database are in a range of formats, including policy-oriented platforms (e.g., the Green Investment Principles for the Belt and Road), business-led coalitions (e.g., Global Energy Interconnection Development and Cooperation Organization), and research-led alliances (e.g., the Belt and Road International Water Alliance).

Compared with the other two governance levels, China's leadership activities at the multilateral level remain much more limited in number and highly selective in geographic scope. Most of these initiatives take the form of partnerships between China and regional organizations, particularly ASEAN, or are embedded within informal multilateral platforms such as the BRICS and the G20. Notable examples include the ASEAN-China Leaders' Joint Statement on Sustainable Development (ASEAN, 2010), G20's Working Group on Sustainable Finance (SFWG, 2021), and the BRICS Initiative on Trade and Investment for Sustainable Development (BRICS China, 2022). These multilateral initiatives are typically issue-driven and flexible in form, emphasizing project-based voluntary cooperation rather than formal regulations. This pattern reveals Beijing's preference for a relational mode of leadership in multilateral arenas through convening and coordination, instead of setting legally-binding agreements.

Geographically, China-led initiatives were initially concentrated in Asia and Europe, but gradually extended to Africa and Latin America (especially since around 2010), and most prominently to a global scope (operating in more than two continents), as the cumulative number of global initiatives more than tripled between 2009 and 2024 (from 15 to 56). This widening spatial footprint reflects not only the diversification of participants in China-led environmental governance initiatives, but also China's growing intent to engage in governance processes that transcend regional boundaries. The increasing prominence of global initiatives indicates that a growing number of Chinese actors began to

influence environmental governance institutions and norms for the whole world.

Fig. 2 maps the temporal evolution of the environmental problems targeted by the initiatives in our database. The most notable growth has been concentrated in the issue of climate change (30%), clean energy (16%), and water-related issue areas (12%). Additionally, the number of initiatives addressing multiple environmental issues (coded as "broad problems") has increased to about 11% by 2024. This trend suggests a growing number of initiatives acknowledge cross-issue synergies and aim to address multiple environmental issues simultaneously. In comparison, other issues such as chemicals and waste, forests, cities, and disaster management have received very little attention from China-led initiatives.

3.2. Governance functions provided by China's leadership activities

We examine the primary governance functions targeted by the China-led initiatives to further unpack the mechanisms through which Chinese actors seek to exercise their environmental leadership. Fig. 3 shows how China-led initiatives engage with a broad spectrum of environmental issues. "Information sharing and dialogue" exists in almost all issue areas except agriculture and food security, and disaster and humanitarian relief. It is also the dominant category in five areas. The dominance of this function suggests that Chinese actors' leadership activities in global environmental governance frequently take the form of facilitating communication, coordination, and building consensus. "Capacity-building and training" also features prominently across many environmental issues, highlighting the emphasis on mutual learning, technical support, and long-term development, which have been central to China's approach to international cooperation, especially South-South cooperation.

In contrast, the provision of funding remains consistently limited across all issues, showing that financial instruments constitute only a minor component of China's environmental leadership strategy. This may be explained by a strong focus on mutual benefits, instead of altruistic aid programmes, in China's international development cooperation (Brautigam, 2011; Dunford, 2020; Yu, 2021). In other words, China's environmental leadership activities have relied more on policy coordination, technical expertise, and knowledge production, rather than being based on direct monetary transfers. Additionally, "rule-making and standard-setting" is highly selective and primarily concentrated in only a few issue areas including agriculture, biodiversity,

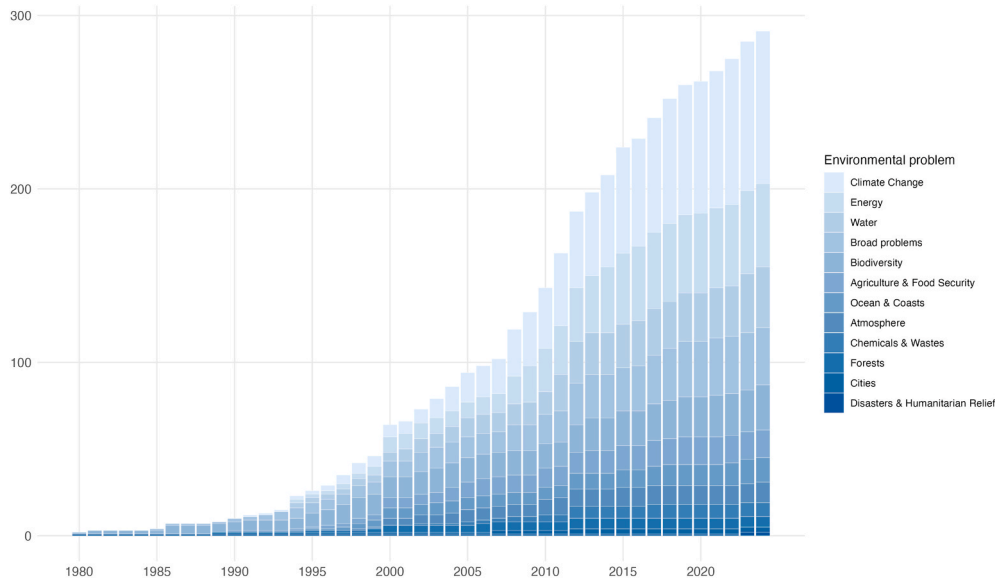


Fig. 2. Environmental problems targeted by these initiatives.

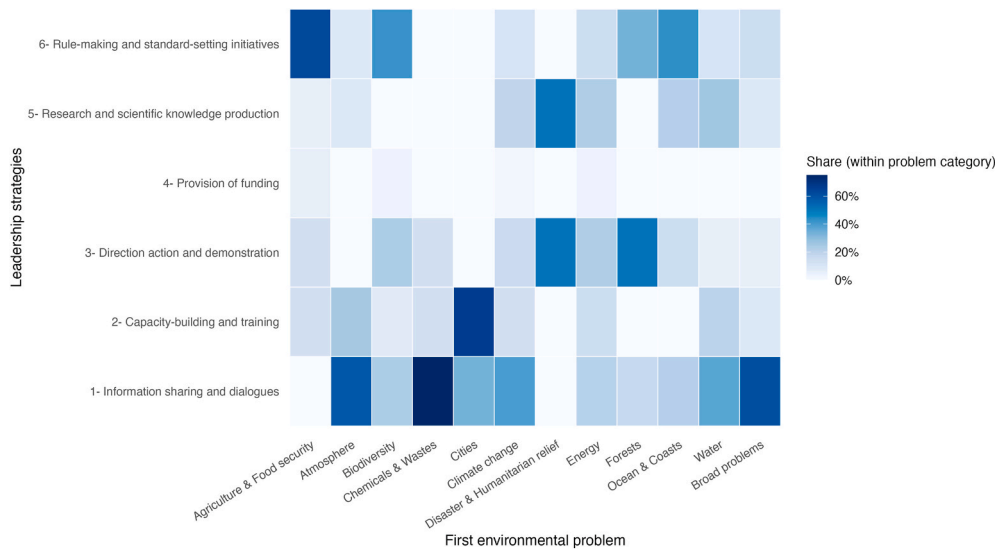


Fig. 3. Governance functions across environmental issue domains.

forests, and oceans – a trend suggesting China’s growing role in shaping global institutions and norms around ecosystem management, resource sustainability, and biodiversity conservation. By contrast, the proportions of rule-making initiatives are much smaller in the areas of climate change, energy, water, and atmosphere where information sharing and dialogue dominates.

Other strategies exhibit a more diffuse pattern. “Direct action and demonstration” appears most prominently in disaster and humanitarian relief, as well as forest governance. In these areas, Chinese actors have sought to project leadership through hands-on engagement, rapid responses, and pilot demonstration projects (Zhu et al., 2024). “Research and scientific knowledge production” is distributed across multiple issues, showing an increasingly active role of Chinese environmental researchers in establishing or shaping international cooperation. This may also reflect the Chinese government’s strategy to use science and technology as a tool of diplomacy (Wagner and Simon, 2023). Finally, it is worth noting that, in our database, climate change and energy initiatives usually have multiple functions, as these issues require a multi-strategy approach due to their inherently complex and cross-sectoral nature.

Fig. 4 shows the variation of Chinese actors’ leadership activities across three governance levels. “Information sharing and dialogues” dominates bilateral initiatives, reflecting the functional and relational logic behind China’s bilateral environmental cooperation where trust-building, technical exchange, and the pursuit of shared development gains are usually prioritized. This approach focuses more on pragmatic knowledge transfer and incremental learning rather than legally binding commitments (Zhu et al., 2024). Compared with other strategies, information sharing also involves lower political and institutional costs, making it a preferred option for sustaining engagement and reducing potential risks in maintaining partnerships. Even as cooperation expands into multilateral arenas, China still emphasizes dialogue- and training-based cooperation for governing common environmental challenges over directive or rule-imposing approaches. For instance, in China-ASEAN cooperative initiatives and China-led projects in BRICS, dialogue and training have served to foster mutual understanding and shared technical practices without directly contesting the existing international governance architecture. In terms of the distribution of governance functions, China’s bilateral and multilateral leadership

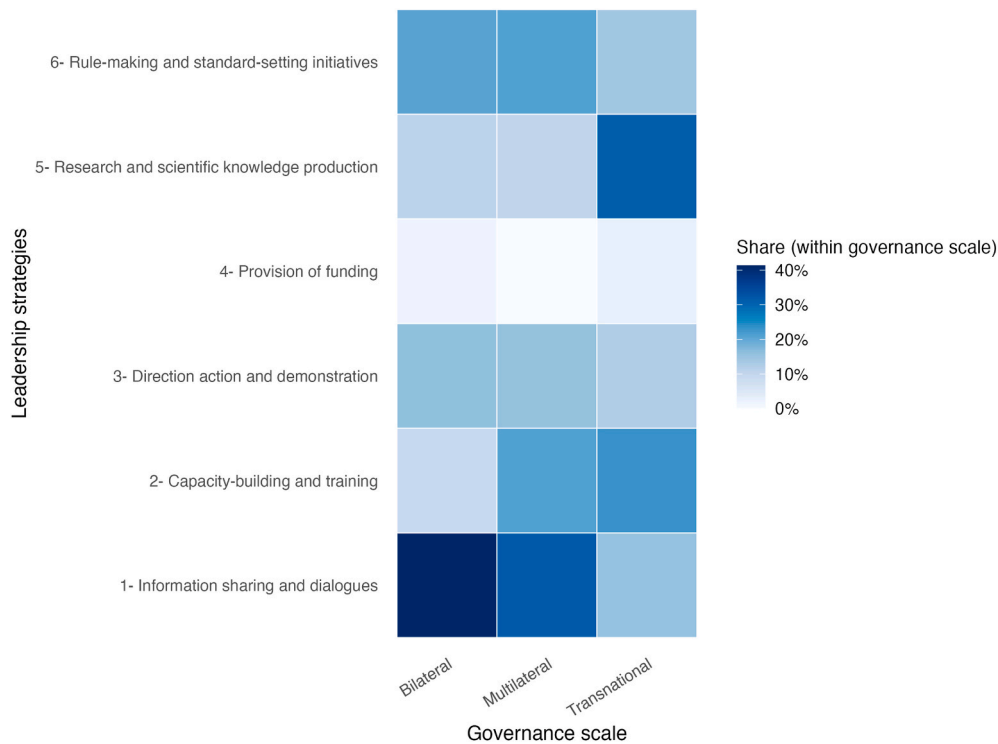


Fig. 4. Variation of leadership activities across governance levels.

activities remain predominantly oriented towards facilitative instruments—instruments that build relationships, share information, and develop technical capacity—rather than instruments designed to impose binding obligations or formal rules.

At the transnational level, Chinese actors' leadership strategies show a different pattern: "research and scientific knowledge production" accounts for approximately 35% of activities, with "capacity-building and training" in second place (roughly 25%). Transnational initiatives often bring together a wide range of non-state actors (e.g., research institutions, businesses, and civil society groups), where their key power resource is derived from knowledge and technical expertise. In such settings, research collaboration allows Chinese actors to frame environmental issues and develop solutions that subtly embed standards, such as those related to clean technologies. Capacity-building functions as an extension of this knowledge production, translating epistemic outputs into actionable governance tools.

This stands in contrast to governance through formal rule-making, illustrating the rise of a distributed, expertise-driven, and network-based governance mode in China's environmental leadership activities in transnational arenas. A typical example is the Third Pole Environment programme established in 2009, led by the Chinese Academy of Sciences, which coordinates international scientific research on the cryosphere and water resources across the Tibetan Plateau and surrounding regions, mobilizing researchers from over twenty countries. In the climate and energy domain, the China–Brazil Center for Climate Change and Energy Technology Innovation, led by Tsinghua University, brings together Chinese and Brazilian researchers to co-develop low-carbon technology pathways (Lewis, 2023b). These initiatives illustrate how Chinese actors increasingly use research collaboration and capacity-building as core mechanisms to promote transnational environmental governance.

It should be noted, however, that this pattern is not unique to China-led initiatives. The transnational environmental governance literature has observed that knowledge production and capacity-building constitute key focuses of many transnational governance networks, especially driven by globalized epistemic communities (Andonova et al., 2009;

Bulkeley et al., 2014; Hale, 2020). The prominence of knowledge-oriented functions at the transnational level in our database may therefore reflect Chinese actors' involvement in this global trend.

3.3. Key actors behind China-led initiatives

Our database also shows that China has partnered with a range of international actors when leading international environmental governance. Fig. 5 illustrates the geographical distribution of the international partners that co-lead the initiatives with Chinese actors. While these partners exist in all world regions, most of them are concentrated in a small number of countries and regions. Interestingly, the US, Europe, and Russia are the top three partners to co-lead international environmental governance with China. Additionally, Chinese actors have co-developed many initiatives with international organizations, such as the United Nations Environment Programme (UNEP), as shown by a relatively high number of co-leads in Kenya.

The prominence of co-leads in the Global North shows a strategy of engaging with actors that have technological advantages and rich experience in environmental governance, allowing for Chinese actors to draw on these partners' resources and expertise when developing new initiatives. This pattern challenges the conventional expectation that frames China's environmental leadership primarily as a South–South phenomenon (Harlan and Lu, 2022; Rodenbiker, 2023). Instead, the database suggests that Chinese actors more frequently lead with Global North-based actors than with their partners in the Global South (see Fig. 5). Cooperation with the US and European countries has focused on joint research, technology demonstration, and the co-development of policy frameworks, such as the US–China Ten-Year Framework on Energy and Environment and the Europe–China Clean Energy Centre (US Department of State, 2009; National Development and Reform Commission of China/European Commission, 2009). In comparison, the partnerships with Russia follow a rather different rationale, with about 90% (17 out of 19) focusing on the management of shared transboundary ecosystems, such as the Protocol on Protection of Tiger, the Agreement on Joint Control of Forest Fire, and the Agreement on

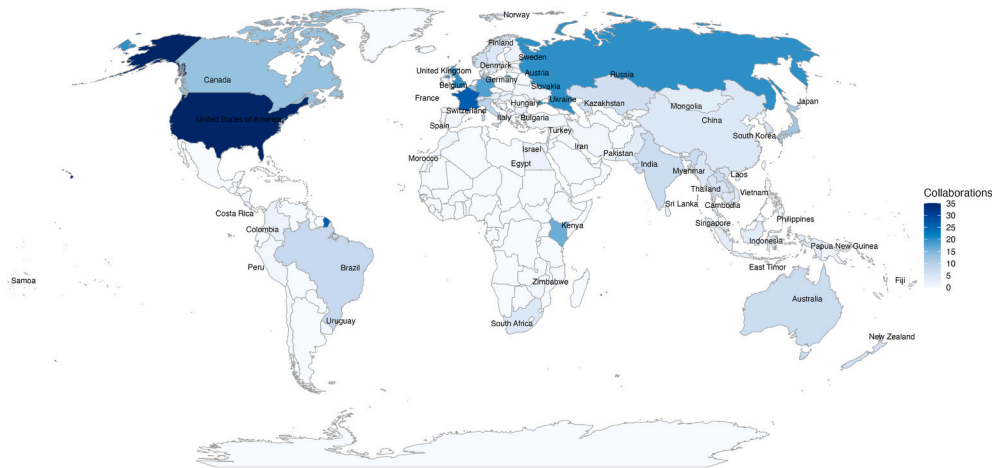


Fig. 5. Geographic distribution of international partners co-leading international environmental governance initiatives with China.

Transboundary Waters.

Differences in colour shades illustrate the level of participation counted by the number of international co-leads, in which darker shades refer to higher levels. For non-state and international organizations, we recorded the countries hosting their headquarters. As a result, some co-lead international organizations, such as Asian Infrastructure Investment Bank, are coded as being from China.

Co-leadership with Global South partners, by contrast, appears to aim at building coalitions to shape agendas in global and regional environmental governance. In the Global South, China's partners are highly concentrated in a few large emerging economies (especially Brazil, India, and South Africa) and Southeast Asia. China's co-leadership activities with other large emerging economies exist across all three governance levels. This trend can be explained by these countries' growing interest in exercising more influence in global environmental governance. As such, building coalitions with these emerging economies – whether bilaterally or in groups like BRICS – can enhance the legitimacy and authority of China's leadership activities. A noteworthy case is the China–Brazil Centre for Climate Change and Energy Technology Innovation, a collaborative platform that demonstrates how

China engages other emerging economies to co-develop climate and energy innovation agendas, thereby strengthening both the perceived legitimacy and the authority of its leadership activities (Lewis, 2023b). Partnerships with Southeast Asian countries are largely anchored in capacity-building, demonstration projects, and policy training. For instance, the Green Lancang-Mekong Initiative, launched in 2015, has established an important platform for multilateral policy dialogues and joint research projects (Junlin et al., 2021). Another important feature in the initiatives co-led by China and its Global South partners is that these initiatives often frame environmental cooperation as part of a shared development trajectory rather than developing regulatory or coordination mechanisms for addressing specific problems.

To better understand the interrelationships of the leading actors in the China-led international environmental governance initiatives, we employed bipartite social network analysis (see Fig. 6). Table 2 lists the top ten organizations ranked by the number of initiatives that each organization (co)leads. Overall, Fig. 6 shows an asymmetrically polycentric, yet partially defragmented, network, in which several Chinese central government agencies are surrounded by diverse semi-peripheries consisting of international and non-state collaborators. The Ministry of

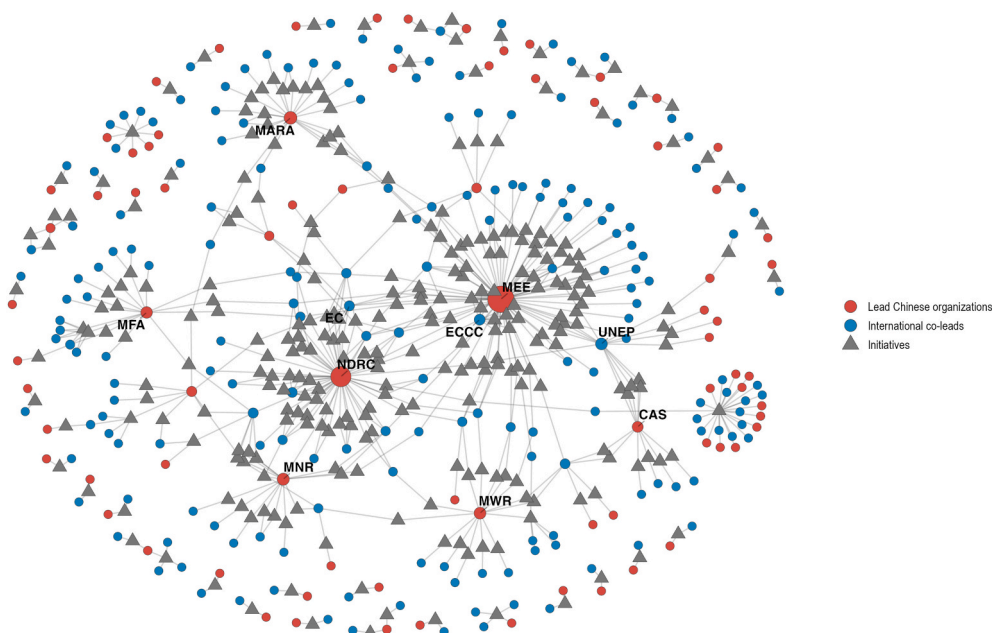


Fig. 6. Bipartite network of China-led global initiatives and leading organizations.

Table 2

Top organizations by participation across China-led environmental initiatives.

Name	Abbreviation	Degree centrality	Page rank
Ministry of Ecology and Environment	MEE	77	0.062
National Development and Reform Commission	NDRC	51	0.026
Ministry of Agriculture and Rural Affairs	MARA	19	0.023
Ministry of Natural Resources	MNR	16	0.013
Ministry of Water Resources	MWR	15	0.018
United Nations Environment Programme	UNEP	15	0.011
Ministry of Foreign Affairs	MFA	14	0.018
Chinese Academy of Sciences	CAS	11	0.009
Environment and Climate Change Canada	ECCC	11	0.003
European Commission	EC	8	0.005

Ecology and Environment (MEE) occupies a central position, linking multiple sub-clusters of initiatives and co-lead organizations. Its high score of degree centrality ($N = 77$) indicates not only institutional prominence but also its role as the central coordination node between domestic actors in China and international partners for establishing international environmental governance initiatives. Around this hub, other Chinese government agencies such as the National Development and Reform Commission (NDRC), the Ministry of Agriculture and Rural Affairs (MARA), the Ministry of Natural Resources (MNR), and the Ministry of Water Resources (MWR) form a multi-core array that distributes functional responsibilities across the issue areas on which they focus.

The grey triangles represent the initiatives, and the circles represent the organizations that jointly lead them. For circles, the red nodes indicate the leading Chinese organizations, and the blue nodes denote their primary international partners. The node size of circles reflects degree centrality, indicating how many initiatives each organization is connected to.

Moreover, the Chinese Academy of Sciences (CAS) has a relatively central position in the network, (co)leading eleven initiatives. CAS is an interesting case to understand the role of other Chinese actors in leading international cooperation. While operating outside the state bureaucracy as a third-party scientific advisory body, it is affiliated with the State Council and receives government funding ((Li et al., 2016)).¹ This status is a characteristic of many Chinese research institutions and industry associations that lead transnational governance initiatives – often in collaboration with Chinese and international non-state actors – in our database. By operating close to, but not fully within, the Chinese state, these actors often have capacity and political influence to lead international cooperative initiatives. This finding echoes the importance of research and scientific knowledge production in China-led transnational governance initiatives, confirming the growing role of knowledge-based institutions in China's leadership activities. These actors with close connections to the state advance China's environmental leadership not through political influence, but through their scientific capacity and convening power.

Additionally, many non-state and subnational actors are located at the periphery to form a series of loose but diverse clusters. For example, the Green Finance Committee of the China Society for Finance and Banking (GFC), as a policy-oriented research platform, has co-led the development of the Green Investment Principles for the BRI and UK-

China Green Finance Centre to promote environmental risk management and sustainable finance (GFP, 2018). The existence of many loosely connected peripheral initiatives like these challenges the conventional view that China's engagement in international environmental cooperation is predominantly monocentric led by the state (Coenen et al., 2021; Lo, 2015). Hence, our findings suggest that multiple centres of authority coexist and interact in China-led international environmental governance initiatives, constituting a polycentric network under state coordination with a range of non-state and subnational actors. The South-South Climate Cooperation Fund, jointly initiated by NDRC and UNDP together with other developing countries, provides a clear example of how authority is dispersed across domestic, international, and intergovernmental actors within a single cooperation framework (Weigel, 2016). Instead of leading to an institutional disorder, this configuration can facilitate integration in which coherence arises from the interconnections among multiple centres of authority (Kim, 2020).

Regarding international co-leads, UNEP, Environment and Climate Change Canada, and the European Commission are the top three organizations in the network (see Fig. 6 and Table 2). This suggests that China's environmental leadership is increasingly co-exercised with other international environmental leaders, including key international organizations and Global North countries with a strong green agenda, to enhance the capacity, credibility, and legitimacy of China-led initiatives. In turn, these partnerships also function as essential bridging nodes, connecting China-led initiatives and related actors with multilateral environmental institutions.

4. Discussion and conclusion

Over the past decade, China's rise as a new global power and its environmental turn have generated a heated debate on the country's role in global environmental governance. Given the US' retreat from environmental action, there are high expectations among researchers and policymakers that China fill a leadership void to advance global sustainability transition. However, until now no comprehensive data exist to track China's evolving influence on global environmental governance. The CGEL Database fills this knowledge gap by systematically mapping China-led international environmental governance initiatives at the bilateral, multilateral, and transnational arenas in the 21st century. Our descriptive analysis shows that China's leadership role has expanded steadily over the past three decades, reflected in the number of new initiatives being developed and their widening geographic reach. While this shift has been mainly coordinated by a few national government agencies, a growing number of Chinese non-state actors have become actively involved in co-developing or co-managing cooperative initiatives. By mapping China's initiatives to lead international cooperation and actors behind them, the database can advance our understanding of China's multi-faceted efforts to shape global environmental governance.

The initiatives in the database also show a strong focus on information sharing and dialogues, capacity-building and training, and research and scientific knowledge production. These broad patterns suggest that China's emerging leadership in global environmental governance relies mainly on the development and dissemination of knowledge and ideas, instead of making and enforcing new international regulations or providing financial support based on its material power. This pattern echoes previous research showing a pragmatic strategy of some Chinese actors to engage with a diverse set of partners, including non-state and intergovernmental organizations to enhance authority, while avoiding developing new rules directly challenging existing institutions and norms led by the Global North (Sun and Yu, 2023; Park, 2023; Zhu et al., 2024). The emphasis on knowledge production and capacity-building also speaks to an alternative conception of international leadership based on the traditional Chinese philosophy, which emphasizes morality and credibility over material dominance (Yan, 2019). While recognizing the politics of knowledge production, future

¹ It remains debatable whether these organizations can be considered as "non-state" actors because the boundaries between state and non-state actors in China are often blurred, and for this reason some researchers describe them as "quasi-state" actors (Guttman et al., 2018; Sun, 2022).

research can focus on relevant cases to further investigate moral foundations and legitimization strategies of China-led initiatives and identify alternative ideas and practices challenging dominant, Eurocentric paradigms on world order and earth system governance (Acharya, 2016; Jayaram et al., 2025). By grounding the question of leadership foundations in the environmental domain, our database can also contribute to the debate on the type of leadership that China seeks to promote in a changing international order (Johnston, 2019; Wallace and Wallace, 2021). Accordingly, the database can help scholars of environmental and sustainability governance further investigate what kind of leader different Chinese actors seek to be in a warming world (Harlan et al., 2025).

In the context of planetary crises, new forms of leadership are emerging and will shape the future of global sustainability transition. By tracking China-led international environmental governance initiatives, the CGEL Database can become the foundation for an important new research agenda on China's shifting role in global environmental governance. Its initiative-level data across multiple dimensions (including lead and partner actors, environmental issues, and governance functions) will allow users to address various research questions based on their interests. Three lines of inquiry are at the core of this agenda. First, future research should further investigate the motivations of different Chinese actors in pursuing international leadership, including economic, political, and normative drivers. Data on lead Chinese actors and their partners also allow researchers to examine how leadership coalitions form and evolve over time. Second, to fully understand China's global influence, environmental governance researchers need to assess the effectiveness of China-led initiatives in governing pressing environmental issues and evaluate the extent to which they fill the gaps left by Northern-led governance arrangements. Users can combine the CGEL Database with geospatial data to conduct impact assessments. Third, the effects of China's environmental leadership on the rest of the world – including both the Global North and South – are worth examining. This requires fine-grained analysis of how China's leadership activities are understood, interpreted, and reacted to by stakeholders from across the globe. Finally, our methodology of tracking China-led initiatives can be also applied to other emerging powers such as BRICS countries to generate comparative knowledge on new forms of leadership in earth system governance.

CRedit authorship contribution statement

Yixian Sun: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Yitong Ye:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.esg.2026.100339>.

Data availability

The CGEL Database is provided as Supplementary Material with this article and is also openly available at <https://doi.org/10.15125/BATH-01664>

References

- Acharya, A., 2016. Advancing global IR: challenges, contentions, and contributions. *Int. Stud. Rev.* 18, 4–15. <https://doi.org/10.1093/isr/viv016>.
- Ahlquist, J.S., Levi, M., 2011. Leadership: what it means, what it does, and what we want to know about it. *Annu. Rev. Polit. Sci.* 14, 1–24. <https://doi.org/10.1146/annurev-polisci-042409-152654>.
- Andonova, Liliana B., Betsill, M.M., Bulkeley, H., 2009. Transnational climate governance. *Glob. Environ. Polit.* 9, 52–73. <https://doi.org/10.1162/glep.2009.9.2.52>.
- ASEAN, 2010. ASEAN-China Leaders Joint Statement on Sustainable Development. Association of Southeast Asian Nations.
- Biermann, F., 2014. Earth system governance: world politics in the anthropocene. *Earth System Governance: a Core Research Project of the International Human Dimensions Programme on Global Environmental Change*. The MIT Press, Cambridge, Massachusetts.
- Bosi-Moreira, B., 2025. A green world order with Chinese characteristics: implications for global climate cooperation. *Glob. Environ. Polit.* 25, 100–117. <https://doi.org/10.1162/glep.a.11>.
- Brautigam, D., 2011. Aid 'with Chinese characteristics': chinese foreign aid and development finance meet the OECD-DAC aid regime. *J. Int. Dev.* 23, 752–764. <https://doi.org/10.1002/jid.1798>.
- BRICS China, 2022. BRICS Initiative on Trade and Investment for Sustainable Development.
- Bulkeley, H., Andonova, L., Betsill, M.M., Compagnon, D., Hale, T., Hoffmann, M.J., Newell, P., Paterson, M., Roger, C., VanDeveer, S.D., 2014. *Transnational Climate Change Governance*, first ed. Cambridge University Press. <https://doi.org/10.1017/CBO9781107706033>.
- Chan, S., Deneault, A., Hale, T., 2022. Climate - Cooperative Initiatives Database (C-CID).
- Coenen, J., Bager, S., Meyfroidt, P., Newig, J., Challies, E., 2021. Environmental governance of China's belt and road initiative. *Environ. Policy Govern.* 31, 3–17. <https://doi.org/10.1002/eet.1901>.
- Doshi, R., 2021. *The Long Game: China's Grand Strategy to Displace American Order, Bridging the Gap*. Oxford University Press, New York. <https://doi.org/10.1093/oso/9780197527917.001.0001>.
- Dunford, M., 2020. Chinese and development assistance committee (DAC) development cooperation and development finance: implications for the BRI and international governance. *Eurasian Geogr. Econ.* 61, 125–136. <https://doi.org/10.1080/15387216.2020.1716821>.
- Feng, H., McCarthy, S., 2025. Partnerships of developmental pragmatism: greenwashing the BRI in southeast Asia. *Environ. Polit.* 1–22. <https://doi.org/10.1080/09644016.2025.2572180>.
- Finamore, B., 2018. *Will China save the Planet?, Environmental Futures*. Polity, Cambridge, UK ; Medford, MA, USA.
- Geng, Q., Lo, K., 2023. China's green belt and road initiative: transnational environmental governance and causal pathways of orchestration. *Environ. Polit.* 32, 1163–1185. <https://doi.org/10.1080/09644016.2022.2156176>.
- GIP, 2018. *Green Investment Principles*.
- Greenstone, M., He, G., Li, S., Zou, E.Y., 2021. China's war on pollution: evidence from the first 5 years. *Rev. Environ. Econ. Pol.* 15, 281–299. <https://doi.org/10.1086/715550>.
- Guttman, D., Young, O., Jing, Y., Bramble, B., Bu, M., Chen, C., Furst, K., Hu, T., Li, Y., Logan, K., Liu, L., Price, L., Spencer, M., Suh, S., Sun, X., Tan, B., Wang, H., Wang, X., Zhang, J., Zhang, X., Zeidan, R., 2018. Environmental governance in China: interactions between the state and “nonstate actors.”. *J. Environ. Manag.* 220, 126–135. <https://doi.org/10.1016/j.jenvman.2018.04.104>.
- Hale, T., 2020. Transnational actors and transnational governance in global environmental politics. *Annu. Rev. Polit. Sci.* 23, 203–220. <https://doi.org/10.1146/annurev-polisci-050718-032644>.
- Hansen, M.H., Li, H., Svarverud, R., 2018. Ecological civilization: interpreting the Chinese past, projecting the global future. *Glob. Environ. Change* 53, 195–203. <https://doi.org/10.1016/j.gloenvcha.2018.09.014>.
- Harlan, T., Sun, Y., Lu, J., DiCarlo, J., Goron, C., Li, Y., Liao, J.C., Mao, K., Rodenbiker, J., Seligsohn, D., Wang, A., Weins, N., Zhu, A.L., 2025. China aspires to be an environmental leader: how should the rest of the world engage? *J. Curr. Chin. Aff.* <https://doi.org/10.1177/18681026251400351>, 18681026251400351.
- Harlan, T., Lu, J., 2022. *Green Cooperation: Environmental Governance and Development Aid on the Belt and Road, 2021–22* Wilson China Fellowship. Wilson Center, Washington, DC.
- Comparative political leadership. In: Helms, L. (Ed.), 2012. *Palgrave Studies in Political Leadership Series*. Palgrave Macmillan, Basingstoke, Hampshire.

- Hilton, I., 2016. China emerges as global climate leader in wake of Trump's triumph. *Guardian*.
- IEA, 2024. *Energy Technology Perspectives 2024*. IEA, Paris.
- Jayaram, D., Inoue, C.Y.A., Kotzé, L.J., Barros-Platiau, A.F., Gonçalves, V.K., Triyanti, A., Balag'kutu, T.A., 2025. Locating the 'Global South' in earth system governance: editorial. *Earth Syst. Govern.* 26, 100301. <https://doi.org/10.1016/j.esg.2025.100301>.
- Johnston, A.I., 2019. China in a world of orders: rethinking compliance and challenge in Beijing's international relations. *Int. Secur.* 44, 9–60. https://doi.org/10.1162/isec_a.00360.
- Junlin, R., Ziqian, P., Xue, P., 2021. New transboundary water resources cooperation for greater mekong subregion: the Lancang-Mekong cooperation. *Water Policy* 23, 684–699. <https://doi.org/10.2166/wp.2021.198>.
- Kim, R.E., 2020. Is global governance fragmented, polycentric, or complex? The state of the art of the network approach. *Int. Stud. Rev.* 22, 903–931. <https://doi.org/10.1093/isr/viz052>.
- Kopra, S., 2020. Leadership in global environmental politics. In: *Oxford Research Encyclopedia of International Studies*. <https://doi.org/10.1093/acrefore/9780190846626.013.537>.
- Lewis, J., 2023. Database of bilateral clean energy and climate agreements with China. <https://doi.org/10.17632/8rzt5ng86.1>.
- Lewis, J.I., 2023. *Cooperating for the Climate: Learning from International Partnerships in China's Clean Energy Sector*. The MIT Press, Cambridge, Massachusetts.
- Li, Xiaoxuan, Yang, Kejia, Xiao, Xiaoxi, 2016. Scientific advice in China: the changing role of the Chinese Academy of Sciences. *Palgrave Communications* 2 (1), 16045. <https://doi.org/10.1057/palcomms.2016.45>. <https://www.nature.com/articles/palcomms201645>. (Accessed 12 July 2016).
- Li, Y., Shapiro, J., 2020. *China Goes Green: Coercive Environmentalism for a Troubled Planet*. Polity, Cambridge, UK ; Medford, MA.
- Lo, K., 2015. How authoritarian is the environmental governance of China? *Environ. Sci. Pol.* 54, 152–159. <https://doi.org/10.1016/j.envsci.2015.06.001>.
- Mazzega, P., Rugmini, D.M., Barros-Platiau, A.F., 2025. Where is the "Global South" located in scientific research? *Earth Syst. Govern.* 25, 100269. <https://doi.org/10.1016/j.esg.2025.100269>.
- Mitchell, R.B., Andonova, L.B., Axelrod, M., Balsiger, J., Bernauer, T., Green, J.F., Hollway, J., Kim, R.E., Morin, J.-F., 2020. What we know (and could know) about international environmental agreements. *Glob. Environ. Polit.* 20, 103–121. https://doi.org/10.1162/glep_a.00544.
- National Development and Reform Commission of China, European Commission, 2009. *Joint Statement on Europe-China Clean Energy Centre*.
- Park, H., 2023. Global norm-maker as China's new brand? An analysis of the responsible cobalt initiative. *Chin. J. Int. Polit.* poad002. <https://doi.org/10.1093/cjip/poad002>.
- Qi, J.J., Dauvergne, P., 2022. China's rising influence on climate governance: forging a path for the global South. *Glob. Environ. Change* 73, 102484. <https://doi.org/10.1016/j.gloenvcha.2022.102484>.
- Rodenbiker, J., 2023. *Ecological civilization goes global: China's green soft power and South-South environmental initiatives*. The Wilson Center.
- SFWG, 2021. *G20 Sustainable Posted on by Finance Working Group*.
- Sun, Y., 2022. Certifying China: the rise and limits of transnational sustainability governance in emerging economies. *Earth System Governance*. MIT Press, Cambridge, MA, USA. <https://doi.org/10.7551/mitpress/14192.001.0001>.
- Sun, Y., Ye, Y., 2025. Global climate governance remains resilient under Trump 2.0. *Glob. Public Policy Gov.* <https://doi.org/10.1007/s43508-025-00131-x>.
- Sun, Y., Yu, B., 2023. Greening China's Belt and road initiative: from norm localization to norm subsidiarity? *Glob. Environ. Polit.* 23, 91–116. https://doi.org/10.1162/glep_a.00685.
- Tobin, P., Huitema, D., Kellner, E., 2024. The empirical realities of polycentric climate governance: introduction to the special issue. *Glob. Environ. Polit.* 24, 1–23. https://doi.org/10.1162/glep_a.00758.
- US Department of State, 2009. *U.S.-China ten-year framework for cooperation on energy and environment* [WWW Document]. URL. <https://2009-2017.state.gov/e/oes/eqt/tenyearframework/index.htm>. (Accessed 1 April 2026).
- Wagner, C.S., Simon, D.F., 2023. China's use of formal science and technology agreements as a tool of diplomacy. *Sci. Publ. Pol.* 50, 807–817. <https://doi.org/10.1093/scipol/scad022>.
- Wallace, J.C., Wallace, J.L., 2021. Domestic politics, China's rise, and the future of the liberal international order. *Int. Organ.* 75, 635–664. <https://doi.org/10.1017/S002081832000048X>.
- Wang, A.L., 2026. *Chinese Global Environmentalism*, first ed. Cambridge University Press. <https://doi.org/10.1017/9781009363976>.
- Watts, J., 2025. 'There is only one player': why China is becoming a world leader in green energy. *Guardian*.
- Weigel, M., 2016. *More Money, More Impact? China's Climate Change South-South Cooperation to Date and Future Trends*. UNDP, China (Research Paper).
- Yan, X., 2019. *Leadership and the Rise of Great Powers, the Princeton-China Series*. Princeton University Press, Princeton.
- Young, O.R., 1991. Political leadership and regime formation: on the development of institutions in international society. *Int. Organ.* 45, 281–308.
- Yu, B., 2021. Cognitive evolution and China's international development cooperation. *Chin. J. Int. Polit.* 14, 481–505. <https://doi.org/10.1093/cjip/poab014>.
- Zhu, A.L., Weins, N., Lu, J., Harlan, T., Qian, J., Barbi Seleguim, F., 2024. China's nature-based solutions in the Global South: evidence from Asia, Africa, and Latin America. *Glob. Environ. Change* 86, 102842. <https://doi.org/10.1016/j.gloenvcha.2024.102842>.